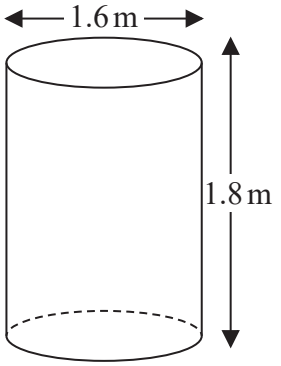
29 Jeremy has to cover 3 tanks completely with paint.

Each tank is in the shape of a cylinder with a top and a bottom.
The tank has a diameter of 1.6 m and a height of 1.8 m.

Jeremy has 7 tins of paint.
Each tin of paint covers 5 m^2

Has Jeremy got enough paint to cover completely the 3 tanks?
You must show how you get your answer.



Radius = 1.6 ÷ 2 = 0.8 m

Two circular ends: $2\pi r^2 = 2\pi \times 0.8^2 = 1.28\pi$ m^2

1 mark
Find the area of the circular ends.

Curved surface = circumference × height

$= 2\pi \times 0.8 \times 1.8 = 2.88\pi$ m^2

1 mark
Find the curved surface area.

One tank: $(1.28 + 2.88)\pi = 4.16\pi$ m^2

Three tanks: $3 \times 4.16\pi = 12.48\pi$

$= 39.207076...$ m^2

1 mark
Find all the area of three tanks.

Paint covers $7 \times 5 = 35$ m^2

1 mark
Find the coverage of seven tins.

No. $39.207... > 35$ m^2,

so Jeremy does not have enough paint.

1 mark
No, with both areas compared.

(Total for Question 29 is 5 marks)